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Branch-Preserving Temporal Travel (BPTT): A Conditional Framework for Source-History-Preserving Backward Temporal Displacement

Branch-Preserving Temporal Travel: A Quantum-Information Framework for Causal History Preservation Under Hypothetical Backward Temporal Displacement

Abstract

Backward time travel creates well-known causal paradoxes, most prominently the grandfather paradox and information or bootstrap paradoxes. A commonly discussed resolution is that a traveller who enters the past does not modify their original history but instead becomes part of an alternative history. This possibility has appeared in philosophical discussions of time travel, branching-spacetime models, and quantum approaches involving closed timelike curves and Everett-style histories.

This paper introduces **Branch-Preserving Temporal Travel (BPTT)** as a formal framework for analyzing this class of scenarios. BPTT does not claim that backward time travel is physically realizable. Instead, it asks a conditional question: **if a physical mechanism permitted backward temporal displacement, could its causal structure be represented as a transition from a source history (A) to a causally distinct destination history (B), while preserving the pre-departure causal history of (A)?**

We formulate a hypothetical branch-transition map (B_{AB}) and define a protected algebra of source-history observables $(A_A^{<t_d})$, where (t_d) is the traveller’s departure event. The central BPTT condition is

$$\mathscr{B}^*_{A\to B}(\mathscr{A}_A^{<t_d}) \subseteq \mathscr{A}_A^{<t_d}$$

$$\mathscr{B}^*_{A\to B}(O_A) = O_A, \quad \forall\, O_A \in \mathscr{A}_A^{<t_d}$$

where (B^{*}) denotes the dual map acting on observables. We show in a simple quantum-information model that this source-history invariance can be implemented consistently when the protected historical degrees of freedom are

separated from the degrees of freedom undergoing state transfer. The construction does not violate unitarity or the no-cloning principle in the toy model.

The framework is then compared with prior proposals, including Lewis's alternate-world resolution of time-travel paradoxes, Deutsch's closed-timelike-curve model, branching-spacetime approaches, and the Entangled Closed Timelike Curve (E-CTC) model of Shoshany and Stober. The comparison indicates that the **core BPTT idea is not novel**: alternate-history resolutions and quantum branching mechanisms are already well represented in the literature. The possible contribution of BPTT is instead its explicit formulation of source-history preservation as an observable-algebra constraint and its separation of ordinary quantum state transfer from the much stronger and currently unestablished notion of controllable branch transfer.

The paper concludes that BPTT should currently be regarded as a **research hypothesis/framework rather than a demonstrated physical theory**. A key open problem is whether its branch-transition map can be derived from existing quantum time-travel models or requires an additional physical postulate.

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1. Introduction

The possibility of travelling through time has occupied both physics and philosophy because backward temporal displacement appears to create contradictions with ordinary causal reasoning. If a traveller were able to return to the past and prevent an event that was necessary for the traveller's own existence, a contradiction would apparently arise.

A common response is to reject the assumption that the traveller must remain on the same history after entering the past. Philosophical discussions of time travel have considered alternate-world and alternate-history resolutions, including the work of David Lewis and later analyses of branching approaches to the grandfather paradox.

Quantum theory has generated additional approaches. Deutsch developed a density-matrix treatment of closed timelike curves (CTCs), imposing a consistency condition on the state of a system interacting with a CTC. His model predicts effects that differ from ordinary quantum evolution, including nonstandard behavior associated with chronology-violating structures.

More recently, Shoshany and Stober proposed **Entangled Closed Timelike Curves (E-CTCs)**, in which parallel timelines are emergent consequences of entanglement between a time-machine system and its environment. Their proposal is explicitly formulated within an Everett or Many-Worlds framework and attempts to provide a concrete quantum mechanism for timelines rather than treating them as informal branches inserted by hand.

The present work begins from a simple conceptual distinction:

$$\text{Forward temporal displacement} \neq \text{Backward temporal displacement}$$

Forward travel into the future is compatible with relativistic time dilation: different observers can accumulate different amounts of proper time. Backward travel, however, raises the additional question of how causal history should be represented if the traveller interacts with the past.

BPTT proposes the following conditional structure:

$$U\_A(t_2) \xrightarrow{T_-} U\_B(t_1), \quad t_1 < t_2, \quad A \neq B$$

The traveller originates in history (A), travels backward, and enters a causally distinct history (B). Events occurring in (B) are therefore not interpreted as rewriting the traveller's original past.

The purpose of this paper is not to claim that this mechanism exists in nature. Instead, the objective is to formulate its causal requirements precisely and determine what can and cannot be represented using ordinary quantum information concepts.

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2. Scope and Scientific Status of BPTT

BPTT is a **conditional hypothesis**.

It begins with the assumption:

> Suppose a physical mechanism capable of backward temporal displacement exists.

The framework then asks:

> What mathematical constraints would be required for that mechanism to preserve the traveller's source history?

This distinction is essential.

The paper does **not** establish:

$$\nexists \ \text{physical time machine} \ \ \text{(not established)}$$

It investigates:

$$\text{If backward temporal displacement exists} \ \Rightarrow \ \text{what causal structure could it have?}$$

The uploaded research review identified this distinction as essential because chronology protection and other general-relativistic considerations challenge the physical realization of closed timelike curves.

Hawking's chronology-protection analysis argues that quantum effects near chronology horizons may prevent the formation of physically realizable closed timelike curves.

Therefore BPTT should not be interpreted as evidence against chronology protection.

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3. Relation to Existing Work

3.1 Alternate-history interpretations

The central BPTT intuition—that travelling into the past may lead to an alternative history rather than rewriting the traveller's original history—is not new.

Lewis's philosophical analysis of time travel discusses alternate-world interpretations of apparent paradoxes, and later literature explicitly identifies branching histories as one possible resolution of the grandfather paradox.

Consequently,

$$\text{past intervention} \ \rightarrow \ \text{alternate history}$$

cannot be claimed as a novel contribution.

3.2 Deutsch's CTC model

Deutsch's 1991 model formulates quantum mechanics near closed timelike lines using a self-consistency condition on the density matrix.

A schematic form of the Deutsch consistency relation is

$$\rho\_CTC = Tr\_CR [U(\rho\_CR \otimes \rho\_CTC)U^\dagger]$$

The defining idea is a fixed point under the CTC evolution.

BPTT differs conceptually because it does not impose the requirement that a traveller's history be globally self-consistent on a single chronology. Instead, it introduces the possibility of a transition to a different history:

$$A \to B$$

The two approaches therefore address paradoxes using different structures:

$$\text{D-CTC: fixed-point consistency}$$

versus

$$\text{BPTT: causal branch separation}$$

However, this distinction alone is not sufficient to establish novelty.

3.3 Branching spacetime

Branching spacetime frameworks have been developed independently of the present work. The research review identified branching and non-Hausdorff spacetime models as important predecessors.

These models demonstrate that a mathematical spacetime structure can contain diverging histories rather than forcing a unique classical continuation through every event.

BPTT therefore does not introduce branching histories as a new ontological concept.

3.4 Entangled Timelines

The closest modern physics comparison is the E-CTC framework of Shoshany and Stober.

They propose a mechanism in which a time-machine system becomes entangled with its environment. As that entanglement spreads, effectively distinct timelines emerge through decoherence. The timelines are therefore not inserted as independent universes from the beginning; they arise as components of the global quantum state.

The E-CTC model uses a unitary evolution (U) and a timeline-correlation operator (T), with a consistency relation of the form

$$TU|\Psi\rangle = e^{i\varphi}|\Psi\rangle$$

The research review identified this model as the closest modern comparison to BPTT.

This comparison is therefore central to the present work.

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4. BPTT Definitions

Definition 1 — Source History

Let (A) denote the traveller's originating causal history.

The physical state of the relevant degrees of freedom is represented by a density operator

$$\rho_A(t)$$

The traveller departs at event

$$p_d = (x_d, t_d)$$

Definition 2 — Destination History

Let (B) denote the causally distinct history reached by the traveller.

The backward transition is written

$$T_- : A(t_d) \to B(t_a)$$

with

[t_a<t_d]

and

$$A \neq B$$

Definition 3 — Departure Time

The departure event defines the boundary between the protected source history and the traveller's subsequent evolution.

The phrase "source-history preservation" therefore refers specifically to the history of (A) **before** (t_d).

This distinction is important because the traveller physically leaving (A) can obviously change the post-departure state of (A).

Definition 4 — Branch-Transition Map

Introduce a hypothetical quantum operation

$$\mathscr{B}_{A\to B}$$

acting on the relevant joint state:

$$\rho' = \mathscr{B}_{A\to B}(\rho)$$

If it represents an ordinary open-system quantum operation, it should be completely positive and trace preserving:

$$\mathscr{B}(\rho) = \Sigma_k K_k \rho K_k^\dagger$$

with

$$\Sigma_k K_k^\dagger K_k = I$$

This does not establish that such a map physically exists. It is the mathematical object whose existence BPTT asks us to investigate.

Definition 5 — Protected Source-Past Algebra

Let

$$\mathscr{A}_A^{<t_d}$$

denote the algebra generated by observables associated with the pre-departure causal history of (A).

An observable

$$O\_A \in \mathscr{A}\_A^\{<t\_d\}$$

represents a measurable property of the source history prior to departure.

For example, a record of a physical event occurring at time ($t<t_d$) may be represented abstractly by ($O_A(t)$).

The traveller’s post-departure degrees of freedom are not automatically included in this algebra.

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5. The BPTT No-Retro-Modification Condition

The central proposed condition is

$$\mathscr{B}^*\_\{A\to B\}(O\_A) = O\_A, \quad \forall\ O\_A \in \mathscr{A}\_A^\{<t\_d\}$$

Here (B^{*}) is the dual map acting on observables.

Using the trace duality relation,

$$\mathrm{Tr}[O\ \mathscr{B}(\rho)] = \mathrm{Tr}[\mathscr{B}^*(O)\rho]$$

the condition implies

$$\mathrm{Tr}[O\_A\ \rho'] = \mathrm{Tr}[O\_A\ \rho]$$

Thus, the expectation value of every protected source-past observable remains unchanged after the branch transition.

This is the mathematical form of the original BPTT intuition:

> Actions performed after arrival in another history cannot rewrite the causal history that existed before the traveller departed the source history.

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6. Why the Earlier Condition (_A'=_A) Is Too Strong

An initial formulation might require

$$\rho\_A' = \rho\_A$$

This is unnecessarily restrictive.

Suppose the traveller is initially part of the source:

$$A = \{\text{environment + traveller}\}$$

After departure,

$$A' = \{\text{environment without traveller}\}$$

The complete post-departure state of (A) therefore need not equal its pre-departure state.

BPTT instead requires preservation of the **historical prefix**:

$$\text{History\_A}(t < t\_d) = \text{History\_A\^\{original\}}(t < t\_d)$$

This distinction makes the framework logically cleaner.

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7. Minimal Quantum-Information Model

Consider three registers:

$$T, A, B$$

where (T) is the traveller's state and (A,B) are abstract destination/source sectors.

Let

$$|\psi\rangle\_T = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

The initial state can be written

$$|\Psi_0\rangle = |\psi\rangle\_T \otimes |0\rangle\_A \otimes |0\rangle\_B$$

Consider a unitary SWAP operation between (T) and (B):

$$U\_TB = I\_A \otimes SWAP\_TB$$

Then

$$U\_TB|\Psi_0\rangle = |0\rangle\_T \otimes |0\rangle\_A \otimes |\psi\rangle\_B$$

Therefore a quantum state can be transferred from one subsystem to another without modifying the third subsystem.

For any observable (O_A) acting exclusively on (A),

$$[U\_TB, O\_A] = 0$$

Consequently,

$$\langle O\_A\rangle' = \mathrm{Tr}[O\_A\ U\_TB\ \rho\ U\_TB\dagger]$$

$$= \mathrm{Tr}[U\_TB\dagger\ O\_A\ U\_TB\ \rho]$$

Thus,

$$\langle O\_A\rangle' = \langle O\_A\rangle$$

This demonstrates mathematical compatibility of source-subsystem invariance with ordinary unitary state transfer.

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8. Important Limitation of the Toy Model

The SWAP construction does **not** demonstrate time travel.

It demonstrates only:

$$\text{quantum state transfer} \neq \text{branch transfer}$$

Registers (A) and (B) are not automatically separate universes or histories.

To interpret them as branches, an additional physical structure is required.

For example, an Everett-style decoherence model may have a global state

$$|\Psi\rangle = \alpha|\psi\_A\rangle|E\_A\rangle + \beta|\psi\_B\rangle|E\_B\rangle$$

with approximately orthogonal environmental records,

$$\langle E\_A|E\_B\rangle \approx 0$$

This produces effectively noninterfering relative histories, but it does not by itself provide a controllable branch-navigation operation.

This distinction is central to the E-CTC literature. Shoshany and Stober describe timelines as emergent decohered components of a global state rather than simply independent universe registers.

9. Source-Past Invariance Under Later Destination Operations

Suppose after the traveller reaches (B), a quantum operation

$$\mathcal{M}\_B$$

acts only within (B).

The final state is

$$\rho\_f = (I\_A \otimes \mathcal{M}\_B)(\rho')$$

For a protected source-past observable (O_A),

$$Tr[O\_A\ \rho\_f] = Tr[O\_A\ \rho']$$

If the branch-transition itself satisfies

$$\mathcal{B}*\_\{A\to B\}(O\_A) = O\_A$$

then

$$Tr[O\_A\ \rho\_f] = Tr[O\_A\ \rho]$$

Therefore, once the protected algebra is invariant under the transition, arbitrary later operations confined to (B) cannot modify it.

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10. Relation to the E-CTC Formalism

The E-CTC framework introduces a global unitary evolution and a timeline-correlation operation satisfying

$$TU|\Psi\rangle = e^\{i\varphi\}|\Psi\rangle$$

The fixed-point condition implies that expectation values of arbitrary global observables are unchanged after a complete consistent cycle:

$$\langle O\rangle\_after = \langle O\rangle\_before$$

However, this is not identical to the BPTT operator condition

$$\mathcal{B}*\_\{A\to B\}(O\_A) = O\_A$$

for an explicitly defined source-history algebra.

For example, in a simplified two-qubit representation where

$$TU$$

acts as a CNOT-like transformation, some operators are invariant while others are transformed. Schematically,

$$(TU)^\dagger Z_1 (TU) = Z_1$$

while

$$(TU)^\dagger X_1 (TU) = X_1X_2$$

Therefore the full operator identity

$$(TU)^\dagger O\_A (TU) = O\_A$$

does not hold for arbitrary source-system observables.

The distinction is important:

$$\text{state-specific fixed-point invariance} \neq \text{universal source-algebra invariance}$$

At the same time, for observables genuinely localized outside the support of a later interaction, ordinary locality can make those observables invariant. Therefore source-history invariance by itself cannot yet be claimed as a new physical law.

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11. No-Cloning Constraint

BPTT requires **transfer rather than duplication**.

The desired transformation is conceptually

$$|\psi\rangle\_A|0\rangle\_B \to |0\rangle\_A|\psi\rangle\_B$$

not

$$|\psi\rangle\_A|0\rangle\_B \to |\psi\rangle\_A|\psi\rangle\_B$$

The latter would constitute cloning of an arbitrary unknown quantum state and is forbidden by ordinary linear quantum mechanics.

Therefore a BPTT mechanism must remove the traveller's physical state from the source trajectory rather than creating an identical second traveller.

Our simple SWAP construction satisfies this requirement.

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12. No-Signalling Constraint

A BPTT model must prevent operations in destination history (B) from becoming a signalling channel into the source history.

For two possible operations ($M_B^{(1)}$) and ($M_B^{(2)}$), require

$$\mathrm{Tr}_B[(I_A \otimes \mathcal{M}_B^{(1)})(\rho)] = \mathrm{Tr}_B[(I_A \otimes \mathcal{M}_B^{(2)})(\rho)]$$

for all source observers restricted to the protected algebra.

Equivalently,

$$P_A(x|\mathcal{M}_B^{(1)}) = P_A(x|\mathcal{M}_B^{(2)})$$

If this condition fails, the model allows observers in (A) to obtain information from choices made in (B), creating a causal communication channel that would require substantially stronger justification.

BPTT therefore adopts

$$B \nrightarrow A$$

as a causal-isolation constraint.

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13. Energy Conservation

Branch transfer introduces a separate physical problem.

Suppose a traveller has total energy (E_T).

Before transfer,

$$E_{total}^{before} = E_A + E_T + E_B + E_M$$

After transfer,

$$E_{total}^{after} = E_A' + E_T' + E_B' + E_M'$$

A physically admissible model must satisfy

$$\Delta E_A + \Delta E_T + \Delta E_B + \Delta E_M = 0$$

where (E_M) represents energy stored in or exchanged with the time-machine mechanism.

The present BPTT framework does not provide a physical Hamiltonian for (B_{AB}), so energy conservation has **not** been derived.

This is a major unresolved requirement.

Existing work on chronology-violating spacetimes also shows that quantum stress-energy effects cannot simply be ignored. Hawking's chronology-protection analysis is a major reason why the physical formation of closed timelike curves remains uncertain.

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14. Causality and the Grandfather Paradox

Consider a traveller originating from

$$A_{2026}$$

Suppose backward travel produces

$$A_{2026} \to B_{2010}$$

The traveller then changes an event in 2010.

BPTT states that this modifies

$$B_{2010} \to B_{2026}$$

not

$$A_{2010} \to A_{2026}$$

Therefore the traveller's original causal history remains intact.

The paradox is replaced by a distinction between histories:

$$A \neq B$$

This is conceptually similar to existing branching-history resolutions, and therefore is not claimed as a novel solution to the grandfather paradox.

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15. Information Bootstrap Problem

Suppose a traveller carries a technical document from 2026 to 2010.

Under naive single-history time travel, the document could create a causal loop:

$$2026 \rightarrow 2010 \rightarrow 2026$$

Its origin becomes ambiguous.

Under BPTT, the document enters (B):

$$A\_2026 \rightarrow B\_2010$$

The document can therefore exist in (B) without becoming a causal prerequisite for its existence in (A).

This removes the simple source-history contradiction, but it does not solve every possible information-loop structure. Additional constraints on the transition map would still be necessary.

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16. Forward/Backward Temporal Asymmetry

One of the original motivations for BPTT is the distinction between future and past travel.

Forward relativistic travel can be represented schematically as

$$U\_A(t_1) \xrightarrow{T_+} U\_A(t_2), \quad t_2 > t_1$$

Backward BPTT travel is instead proposed as

$$U\_A(t_2) \xrightarrow{T_-} U\_B(t_1), \quad t_1 < t_2$$

Thus,

$$T_+ \neq T_-$$

At present, however, this is a **postulate**, not a derived result.

A successful future version of BPTT would need to derive this asymmetry from deeper principles, such as decoherence, causal structure, boundary conditions, or a more fundamental theory.

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17. Mathematical Consistency Assessment

The current framework can therefore be summarized as follows.

Requirement	Current status
Abstract quantum state transfer	Consistent
Unitary toy-model construction	Consistent
No-cloning	Consistent in transfer formulation
Protected source-past observables	Mathematically definable
No-signalling	Can be imposed consistently
Causal-history preservation	Consistent as a constraint
Physical branch-transition mechanism	Not derived
Energy conservation	Not yet demonstrated
Backward temporal transport	Assumed conditionally
New experimentally distinguishable prediction	Not established

The uploaded research plan reached a similar intermediate conclusion: branch-transition construction, energy accounting, falsifiability, and comparison with entangled-timeline models remain the main unresolved objectives.

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18. Novelty Assessment

The literature review leads to a deliberately conservative conclusion.

The following elements are **not novel by themselves**:

alternate history

branching solution to grandfather paradox

Many-Worlds interpretation

quantum closed timelike curves

decoherence-generated timelines

These already occur in prior philosophical, mathematical, and quantum approaches. The research review specifically identified Lewis, Deutsch, branching-spacetime work, and Shoshany–Stober as major overlaps.

The potential contribution of BPTT is narrower:

$$\text{formal branch-transition framework constrained by a protected source-history algebra}$$

However, even this cannot yet be claimed as a unique theorem because related invariance can follow from ordinary locality and decoherence.

Therefore the appropriate novelty statement is:

> **BPTT provides a proposed formalization and comparative framework for source-history-preserving backward temporal displacement. Whether this framework constitutes a genuinely new physical theory remains unresolved.**

This is considerably more defensible than claiming that branching time travel itself is a new idea.

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19. Falsifiability

A major problem remains: if the destination branch is permanently causally disconnected from the source branch, many consequences may be unobservable from the source branch.

For example,

$$A \to B$$

followed by a change in B may produce no observable change in A.

In that case,

$$P\_A(X|BPTT) = P\_A(X|\text{no backward influence})$$

If this equality holds for every accessible observable, BPTT may be empirically indistinguishable from a theory in which backward travel simply does not occur.

This is a serious scientific limitation.

The research review identified exactly this issue: many branch-time-travel models produce null consequences within a source branch and therefore face a difficult falsifiability problem.

A future BPTT model therefore needs either:

1. a measurable energy signature,
2. an observable effect associated with branch transfer,
3. a new quantum-information effect,
4. or a mathematically distinguishable prediction relative to existing E-CTC models.

Without such a difference, BPTT remains primarily a conceptual framework.

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20. Limitations

Several limitations must be made explicit.

First, no physical mechanism for backward temporal displacement is derived.

Second, the branch-transition map

$$\mathscr{B}\_\{A \to B\}$$

is currently an assumed mathematical object rather than a derived interaction.

Third, the energy bookkeeping of transferring a macroscopic physical system between histories has not been solved.

Fourth, the relationship between BPTT and the exact relative-state construction of E-CTCs requires further formal analysis.

Fifth, the current framework does not establish an experimentally accessible prediction that distinguishes BPTT from all existing alternatives.

Finally, chronology protection remains an important obstacle to the physical implementation of backward time travel.

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21. Proposed Future Work

The most important next problem is to determine whether the BPTT transition map can be derived from an existing E-CTC construction.

Specifically, given the E-CTC consistency relation

$$TU|\Psi\rangle = e^{i\varphi}|\Psi\rangle$$

we seek a formally defined relative-state decomposition

$$|\Psi\rangle = \Sigma_i \alpha_i |\psi_i\rangle|E_i\rangle$$

such that a source branch (A) and destination branch (B) can be identified.

The key question becomes:

$$A(t_d) \to B(t_a), \quad t_a < t_d$$

while preserving the source-history prefix?

If the answer is yes, BPTT is best interpreted as a reformulation of existing E-CTC physics.

If the answer is no, one may investigate whether an additional transition structure can be constructed without violating quantum information principles.

A successful extension would need to establish:

$$\mathscr{B}_{A\to B}$$

together with:

$$\mathscr{B}^*_{A\to B}(O_A) = O_A$$

complete positivity,

$$\mathrm{Tr}[\mathscr{B}(\rho)] = 1$$

appropriate energy conservation,

$$\Delta E_{total} = 0$$

and no-signalling from the destination history to the protected source history.

Core Mathematical Conditions

$$B \nrightarrow A$$

$$\Delta E_{total} = 0$$

$$\mathscr{B} \text{ is CPTP: } \mathscr{B}(\rho)=\Sigma_k K_k\rho K_k^\dagger, \quad \Sigma_k K_k^\dagger K_k=I$$

$$\mathrm{Tr}[O_A \rho'] = \mathrm{Tr}[O_A \rho]$$

$$\mathscr{B}^*_{A\to B}(O_A) = O_A, \quad \forall O_A \in \mathscr{A}_A^{<t_d}$$

$$\mathscr{B}^*_{A\to B}(\mathscr{A}_A^{<t_d}) \subseteq \mathscr{A}_A^{<t_d}$$

$$\mathscr{B}_{A\to B}: \rho \to \rho'$$

Contribution and Scientific Status

• BPTT is proposed as a conditional formal framework, not as evidence that physical backward time travel exists.

• The central formal requirement is source-history invariance: $B^*_{A\to B}(O_A)=O_A$ for every protected pre-departure observable O_A.

• The manuscript distinguishes ordinary quantum state transfer from the stronger, currently unestablished concept of controllable transfer between decohered histories.

• The manuscript does not claim that branching time travel, CTC paradox resolution, or Many-Worlds branching are new ideas; those have substantial prior literature.

• The remaining open problem is physical: derive a branch-transition mechanism and a prediction that cannot be reproduced by existing quantum/CTC frameworks.

22. Conclusion

A decisive future result would require a physically derived transition mechanism together with an experimentally distinguishable prediction that cannot be reproduced by ordinary forward-time quantum dynamics or existing CTC/E-CTC constructions.

The contribution is deliberately limited: the paper provides a common mathematical language for comparing these ideas, separates branch transition from ordinary quantum state transfer, and identifies the physical assumptions that would have to be added before BPTT could become a testable new-physics theory.

The formal condition $B^*_{A\to B}(O_A)=O_A$ is useful as a precise statement of source-history preservation, but the manuscript does not claim that this condition is a new physical law. Related ideas concerning alternate histories, closed timelike curves, decoherence, and entangled timelines already exist.

BPTT is best understood at present as a conditional phenomenological and mathematical framework. It formalizes a hypothetical scenario in which a traveller departing from history A at t_d undergoes backward temporal displacement and enters a causally distinct history B at $t_a < t_d$, while protected pre-departure observables of A remain invariant.

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Author’s note on status

This manuscript deliberately distinguishes **known results**, **mathematical constructions**, **assumptions**, and **open hypotheses**. No claim is made that BPTT demonstrates an experimentally realizable time machine, that branch-to-branch travel has been observed, or that the proposed framework has already been established as a novel physical theory.

The next required research stage is an equation-level comparison between the BPTT transition structure and the full relative-state/E-CTC formalism, followed by a rigorous search for an experimentally distinguishable prediction.